

# GUIDELINES for the Launcher Design & Analysis Report

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## Design approach:

1. Choose a stage (lower, upper...): Thrust Range → Thrust value
2. Choose an engine cycle (Staged Combustion, Gas Generator, Expander)
3. Choose a propellant combination
4. Perform the engine design with a step-by-step approach:
  - a. Given the thrust → parametric analysis for optimum Pressure / MR
  - b. Design of the Combustion Chamber
  - c. Design the Combustion Chamber with Cooling Jacket (optional, pending the cycle)
  - d. Oxidizer Pump design
  - e. Fuel Pump design
  - f. Gas Generator/Pre-burner design (if any)
  - g. Turbopump subsystem design
5. Harmonize the interface/Boundary conditions between the subsystems
6. Integrate the subsystems
7. Cross-check and harmonize the initial conditions (mass flows, pressure chains...)
8. Perform design simulation
9. Parametric analysis:
  - a. What if you change the  $P_c$ ?
  - b. What if you change the MR?
  - c. What if you change the Turbopump efficiency(ies)

## Report approach:

1. Provide the mission requirements & the engineering assumptions (initial design choices...) and the main system parameters
2. Describe the combustion chamber design approach (design choices, assumptions, tables, plots results..) Justify each design choice:
  - a. combustion chamber performances ( $P_c$ , MR,  $I_{sp}$ , Thrust)
  - b. combustion chamber contour (convergent, divergent)
  - c. Injector plate pressure drops
  - d. cooling jacket design, contour, material, DP, Max  $T_{wall}$ ...
3. Describe Pump design:
  - a. initial assumptions  $N_s$ , efficiency
4. Describe Turbine design:
  - a. initial assumptions:  $\omega$ , max tip velocity, Mach, Pressure ratio...
5. Repeat step (2) in case of Pre-Burner/ Gas-Generator

6. Describe and present evidence of your step-by-step approach: describe how you integrate together the different subsystems
7. Show the system results:
  - a. Main performances of the Engine
  - b. Main variables/parameter for each component/subsystem
  - c. Differences between simulation output and initial assumptions
8. Comment, justify your results

Provide a table for each component (design assumptions, input to the component, output). Justify the main design choices (throat diameter, chamber pressure, MR, pressure drop in the injectors, pressure drop in the lines, efficiencies, Ns, operative mode of the TurboPumps (TPs)...) )

Provide graphical and numerical evidence of your calculations/simulations (graphs, plots, pictures, figures....)

## Transient Analysis

In case the steady state design has been accomplished, a new task can be pursued.

1. Rebuild your system using the ESPSS transient components.
2. Check if similar steady state results are obtained (be aware that the mathematical models are different; different results are expected)
3. Implement reasonable initial conditions in terms of Pressure and Temperature in your system:
  - a. where are the propellants at TIME = 0 ?
  - b. at which conditions they are at TIME = 0 in each part of the engine?
4. Implement a valve opening sequence in order to achieve a smooth thrust build-up and a not-risky transient start-up of the engine

## During design procedure

IF a NOK appears during compilation check possible mistakes in the schematics:

- Did you properly connect the components?
- Did you declare in a proper way your user given variables?
- Did you provide unphysical/unrealistic values in the components?

IF the Partition creates an experiment with not-wanted/not-expected boundaries please check the different flags present in each components.

## Typical Value ranges for the main parameters (assuming H2/O2):

Inlet LH2:

Pressure domain: 2.80 ; 3.3 bar

Temperature domain: 20 ; 23.6 K

Inlet LO2:

Pressure domain: 2.00 ; 2.3 bar

Temperature domain: 90 ; 92.7 K

Pump efficiency: 0.5 ; 0.7

Turbine efficiency: 0.5 ; 0.7

Combustion efficiency: 0.97 ; 0.99

Pump Ns: 10 ; 20

GasGenerator/Preburner MR: 0.6 ; 0.9

Pump Lox Omega < Pump LH2 omega

Lines Pressure Drop 1.5 ; 5 bar

Maximum wall Temperature (Copper) 670 K